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Sample Test 1

ACT Assessment®

Directions

This booklet contains tests in English, Mathematics, Reading, and Science. These tests measure skills and abilities highly related to high school course work and success in college. *CALCULATORS MAY BE USED ON THE MATHEMATICS TEST ONLY.*

The questions in each test are numbered, and the suggested answers for each question are lettered. On the answer document, the rows of ovals are numbered to match the questions, and the ovals in each row are lettered to correspond to the suggested answers.

For each question, first decide which answer is best. Next, locate on the answer document the row of ovals numbered the same as the question. Then, locate the oval in that row lettered the same as your answer. Finally, fill in the oval completely. Use a soft lead pencil and make your marks heavy and black. *DO NOT USE A BALLPOINT PEN.*

Mark only one answer to each question. If you change your mind about an answer, erase your first mark thoroughly before marking your new answer. For each question, make certain that you mark in the row of ovals with the same number as the question.

Only responses marked on your answer document will be scored. Your score on each test will be based only on the number of questions you answer correctly during the time allowed for that test. You will NOT be penalized for guessing. *IT IS TO YOUR ADVANTAGE TO ANSWER EVERY QUESTION EVEN IF YOU MUST GUESS.*

You may work on each test ONLY when your test supervisor tells you to do so. If you finish a test before time is called for that test, you should use the time remaining to reconsider questions you are uncertain about in that test. You may NOT look back to a test on which time has already been called, and you may NOT go ahead to another test. To do so will disqualify you from the examination.

Lay your pencil down immediately when time is called at the end of each test. You may NOT for any reason fill in or alter ovals for a test after time is called for that test. To do so will disqualify you from the examination.

Do not fold or tear the pages of your test booklet.

**DO NOT OPEN THIS BOOKLET
UNTIL TOLD TO DO SO.**



ENGLISH TEST

45 Minutes—75 Questions

DIRECTIONS: In the five passages that follow, certain words and phrases are underlined and numbered. In the right-hand column, you will find alternatives for each underlined part. You are to choose the one that best expresses the idea, makes the statement appropriate for standard written English, or is worded most consistently with the style and tone of the passage as a whole. If you think the original version is best, choose "NO CHANGE."

You will also find questions about a section of the passage, or about the passage as a whole. These questions

do not refer to an underlined portion of the passage, but rather are identified by a number or numbers in a box.

For each question, choose the alternative you consider best and fill in the corresponding oval on your answer document. Read each passage through once before you begin to answer the questions that accompany it. You cannot determine most answers without reading several sentences beyond the question. Be sure that you have read far enough ahead each time you choose an alternative.

PASSAGE I

Diane Boyd, Wildlife Biologist

[1]

They call her Diane of the Wolves. Wildlife¹
biologist Diane Boyd studies wolves in their native
habitat in a remote wilderness in the Northwest.

It being her base, Boyd's eighty-year-old cabin is
thirty-five miles from the nearest paved road.² It has

little running water, electricity, or phone, and³
Glacier National Park is just across the river that

flows past her cabin door.⁴

[2]

Because of her isolation, Boyd has had to
become self-sufficient. She heats the cabin with⁵

firewood, she chops herself, and hunts wild game⁶
for food. Much of her time is spent trapping wolves
in order to radio-collar (put a tracking device on)
them. Once a wolf is collared, its movements are⁷
easy to follow and record. And though she often
comes across dangerous grizzlies, Boyd carries no
weapon as she makes the rounds of her traps.

1. A. NO CHANGE
B. Wolves, wildlife
C. Wolves. Wildlife,
D. Wolves, specializing in wildlife
2. F. NO CHANGE
G. Her base is an eighty-year-old cabin
H. Being an eighty-year-old cabin, it is her base
J. Boyd's cabin, it being an eighty-year-old base, is
3. A. NO CHANGE
B. less
C. few
D. no
4. F. NO CHANGE
G. passed her cabin
H. passed her cabin's
J. past her cabins'
5. A. NO CHANGE
B. cabin which is her base
C. cabin, her home base
D. cabin, which is eighty years old
6. F. NO CHANGE
G. firewood, she chops herself
H. firewood she chops, herself
J. firewood she chops herself
7. A. NO CHANGE
B. whereby its
C. so that
D. their

GO ON TO THE NEXT PAGE.



While checking her traps, she packs supplies.

8

[3]

[1] The first wolf Boyd collared was Sage, a big gray wolf so strong he has crossed the Continental Divide twice in the dead of winter.

[2] One winter afternoon, on a routine flight to check on her wolves, Boyd spotted Sage in a hunter's trap. [3] She knew that if the hunter didn't kill him, freezing or starvation soon would. [4] If Sage died, Boyd's research on this animal would end unhappily after only four years, it was after landing,

9

Boyd and a fellow, biologist drove fifty miles in a four-wheel-drive pickup. [5] They then traveled through deep snow on snowmobiles in order to reach Sage. [11]

[4]

[1] They approached the half-frozen wolf, sedated him, and wrapped him in a sleeping bag. [2] The sky was dark by the time the two biologists had began working. [3] They used the warmth of their bare hands to restore circulation to Sage's frozen right foot. [4] Just before daylight, they fitted the recovered wolf with a new radio collar and set him free. [13]

[5]

A few days later, Boyd noticed Sage and five of his pup's playful wrestling on the side of a

14

8. Which of the alternatives best provides new, specific details about the equipment Diane Boyd uses for the purpose of trapping and collaring wolves?

- F. NO CHANGE
- G. To manage the wolves, she packs a four-foot pole, tranquilizers, and a hypodermic needle.
- H. Safely collaring a wolf requires equipment, which she brings with her when she checks her traps.
- J. Although she doesn't bring weapons, she arms herself with equipment.

9. A. NO CHANGE
B. years so then after
C. years. After
D. years, after

10. F. NO CHANGE
G. fellow biologist, drove fifty miles
H. fellow biologist drove fifty miles
J. fellow biologist, drove fifty miles,

11. The writer wants to add the following explanation of Sage's name to Paragraph 3:

Boyd named the wolf for his air of wisdom and experience.

This sentence would most logically be placed:

- A. before Sentence 1.
 - B. after Sentence 1.
 - C. after Sentence 2.
 - D. after Sentence 5.
12. F. NO CHANGE
G. have begun
H. begun
J. began
13. Which of the following sequences of sentences will make Paragraph 4 most logical?
- A. 1, 2, 4, 3
 - B. 1, 4, 3, 2
 - C. 2, 1, 3, 4
 - D. 2, 3, 1, 4
14. F. NO CHANGE
G. pup's playfully wrestling
H. pups' wrestling playfully
J. pups wrestling playfully

GO ON TO THE NEXT PAGE.

mountain. Sage's rescue had been a success after all!

15

15. The writer wants to link the essay's opening and concluding sentences. Which of the alternatives most successfully achieves this effect?

A. NO CHANGE
 B. success even so.
 C. success, which was predictable.
 D. success, and Diane of the Wolves had earned her nickname once again.

PASSAGE II

World Trade: Lost in Translation?

As American businesses explore overseas markets. They learn firsthand how language 16 differences can stand in the way of trade. After 17 experiencing a period of slow gasoline sales in a new foreign market, an American oil company learned that its name in the foreign nation's language means "stalled car." A major American car 18 manufacturer, you see, found out that the name of one of its models being converted into hard cash 19 in South America is a Spanish word meaning "ugly old woman." Blunders like these illustrate that an important step in breaking down barriers to international trade is to break down language barriers.

The many countries of Western Europe have always faced this problem. Overcoming it is one of the tasks of the European Economic Community or Common Market an organization 20 founded at its start to promote trade among 21 nations in that part of the world. But it's a difficult

16. F. NO CHANGE
 G. markets, they
 H. markets; they
 J. markets and
17. A. NO CHANGE
 B. differences, which
 C. differences that
 D. differences of which
18. F. NO CHANGE
 G. manufacturer, as proof,
 H. manufacturer
 J. manufacturer, consequently,
19. A. NO CHANGE
 B. sold over the counter
 C. traded for the local currency
 D. offered for sale
20. F. NO CHANGE
 G. Community, or Common Market;
 H. Community, or Common Market,
 J. Community or, Common Market,
21. A. NO CHANGE
 B. which was begun and founded
 C. that it organized
 D. formed

GO ON TO THE NEXT PAGE.

task. When one of the European Economic Community's (EEC's) twelve official languages are used at a meeting, translation must be available

22

to the other delegates. ²³ Finding translators for English, French, and German is relatively easy. But finding them for languages like Danish, Portuguese, and, especially, Greek has proven

more difficult. In fact, when it was

24

almost two decades ago becoming clear that Greece would join the Common Market, officials began to look for translators three years in advance. The results of their search were disappointing to this day, it is almost impossible for them to find translators who can turn Greek

26

into, say, Danish.

27

Then, a standing joke among language officials at EEC headquarters begins

28

with the interrogation of someone by asking the question "What is a Great Dane?" With a rueful smile, someone is likely to respond, "Any Dane who knows how to speak Greek?" All that's needed

29

22. F. NO CHANGE
G. is used
H. are being used
J. are in use

23. At this point, the writer is considering the addition of the following sentence:

Brussels, the home of EEC headquarters, is the lively and modern capital of Belgium.

Would this be a logical and relevant addition to the essay?

- A. Yes, because it serves to establish the setting for the essay.
B. Yes, because it helps to legitimize the EEC by mentioning its main headquarters.
C. No, because the official languages of Belgium aren't mentioned as "problem" languages.
D. No, because it sheds no new light on the problem of language barriers in the EEC.

24. F. NO CHANGE
G. (Do NOT begin new paragraph) Furthermore,
H. (Begin new paragraph) In fact,
J. (Begin new paragraph) Furthermore,

25. A. NO CHANGE
B. (Place after *clear*)
C. (Place after *join*)
D. (Place after *began*)

26. F. NO CHANGE
G. disappointing, and
H. disappointing and
J. so disappointing

27. A. NO CHANGE
B. into say,
C. into, say
D. into,

28. F. NO CHANGE
G. A
H. For instance, a
J. Eventually, a

29. A. NO CHANGE
B. by asking someone to answer the
C. by commencing with the introductory
D. with the

GO ON TO THE NEXT PAGE.

to turn that smile into a look of despair, is a
 reminder that Turkey, whose language is not
 widely studied in Western Europe, may someday
 become a member of the Common Market.

30. F. NO CHANGE
 G. despair, was
 H. despair is
 J. despair, would be

PASSAGE III

Central Park

[1]

While New Yorker's sometimes take Central
 Park for granted, visitors are often astonished to
 discover the size and variety. Nestled in the very
 center of the nation's largest city are 843 acres of
 wooded and landscaped grounds containing lakes
 and ponds, bogs and meadows, a castle, a zoo, and
 an enormous range of wildlife. Though it seems to
 have always been there, Central Park represents
 only the most recent in a series of altering
 transformations of this part of the country.

[2]

[1] About 450 million years ago,
what is now park rested on the floor of an ancient
 sea. [2] After aeons of erosion, accelerated by the
 Ice Age and the landscape began to assume its
 modern appearance. [3] By the time European
 settlers arrived, the entire island was an oak-and-
 chestnut woodland, it is populated by black bears,
 wolves, and beavers. [4] Within the next 200 years,
 however, the forests were cleared for farming, and
 the animals were eliminated by hunters and trappers.

31. A. NO CHANGE
 B. Yorker's,
 C. Yorkers
 D. Yorkers'
32. F. NO CHANGE
 G. its
 H. it's
 J. their
33. A. NO CHANGE
 B. changes,
 C. changing
 D. OMIT the underlined portion.
34. At this point, the writer would like to provide specific
 geographical information about the area where Central
 Park is located. Which alternative does that best?
 F. NO CHANGE
 G. these parts.
 H. this part of the island of Manhattan.
 J. this part that resulted in Central Park.
35. A. NO CHANGE
 B. that which is now park, but was not then
 C. that park, which then was not one,
 D. it wasn't a park then, but it
36. F. NO CHANGE
 G. Age;
 H. Age,
 J. Age, and
37. A. NO CHANGE
 B. it was
 C. they found it
 D. OMIT the underlined portion.

GO ON TO THE NEXT PAGE.



[5] By the mid-nineteenth century, central

Manhattan, being barren of trees, creating a
³⁸
desolate area of swamps, hovels, and pigsties.

[6] Over 200 million years ago, the shifting of the
earth's surface squeezed present-day Manhattan up
into mountains over 12,000 feet high. 39

[3]

In 1857 Frederick Olmsted and Calvert

Vaux were given the enormous task of transforming
⁴⁰
this area into a park. Their intention was

to finish the park within two years. They undertook
⁴¹

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the project with confidence and enterprise, overturning
⁴²
nearly five million cubic yards of rock and earth.

Olmsted and Vaux hired local land developers
⁴³
to help plan the park's landscape.
⁴³

[4]

Today in Central Park, natural beauty and
human engineering work together. Although the
⁴⁴

genius of Olmsted and Vaux, once a landscape barely
⁴⁵

38. F. NO CHANGE
G. Manhattan being
H. Manhattan, which was
J. Manhattan was

39. For the sake of logic and coherence, Sentence 6 should be placed:
A. where it is now.
B. before Sentence 1.
C. before Sentence 2.
D. before Sentence 5.

40. F. NO CHANGE
G. were gave
H. was given
J. been given

41. At this point, the writer wants to provide readers with a general description of Vaux and Olmsted's vision of the finished park. Assuming all are true, which alternative does that best?
A. NO CHANGE
B. to find new techniques for land development by experimenting with new machinery and new ideas in land design.
C. to create the park using the most economical means they could.
D. not to create a formal garden with meticulously trimmed hedges, as in the European tradition, but a natural-looking landscape.

42. F. NO CHANGE
G. they overturn
H. they overturned
J. they having overturned

43. At this point, the writer would like to emphasize the enormousness of the project. Assuming all are true, which alternative does that best?
A. NO CHANGE
B. Another 700,000 cubic yards of topsoil had to be brought from New Jersey before the landscapers could plant their 17,000 trees and shrubs.
C. The result is a park that attracts tourists from all over the world.
D. Many workers from the New York area contributed to the park's development.

44. F. NO CHANGE
G. Considering
H. Through
J. Despite

45. A. NO CHANGE
B. a once-bare landscape
C. a once barely landscape
D. a once, bare landscape

GO ON TO THE NEXT PAGE.

was transformed into an oasis in the center of urban life.

Item 46 poses a question about the essay as a whole.

46. Suppose the writer were to eliminate Paragraph 1. This omission would cause the essay as a whole to lose primarily:
- F. relevant details about the current attractions in Central Park.
 - G. irrelevant details about the current attractions in Central Park.
 - H. historical information regarding the creation of Central Park.
 - J. an irrelevant anecdote about the tourists in Central Park.

PASSAGE IV

My Most Memorable Meal

[1]

Recently I came across a series of magazine articles which various people described their most

47

memorable eating experiences. Out of all those people who were interviewed many of those people

48

raved about the perfect seven-course meal they had once been served in a grand and expensive restaurant. Some reminisced about romantic picnics. Meals made memorable by the

50

diner's companions; rather than by the

51

47. A. NO CHANGE
B. that
C. in which
D. OMIT the underlined portion.
48. F. NO CHANGE
G. Out of all the ones they interviewed, many of those people
H. Of the ones that were interviewed, out of those, many of them
J. Many of those interviewed
49. A. NO CHANGE
B. meals, that
C. meals, which
D. meals,
50. F. NO CHANGE
G. picnics; meals
H. picnics—meals
J. picnics meals
51. A. NO CHANGE
B. diners' companions
C. diners companions'
D. diners companions

GO ON TO THE NEXT PAGE.

food consumed. 52

[2]

While reading these entertaining memoirs, I was myself moved to rummage through old memories in search of remarkable meals. My first meal at the college dormitory came to mind, (the name of it was Goodhue) but it was hardly ⁵³ memorable for its excellence. I considered the string of Thanksgiving dinners I've enjoyed, but, and perhaps by design, no one of them stand ⁵⁴

out as unusual. 55

[3]

Then I remembered the night I tasted the Sri Lankan holiday dish called *lampries*. Served in a rough cottage in the Sri Lankan jungle, with lizards climbing the whitewashed walls and the innkeeper's dog leaping affectionately into my lap, the meal was certainly not remarkable for its formal elegance. It was, however, the more ⁵⁶ complex, elaborate, and delicious concoction I've ever eaten.

[4]

It took the innkeeper and the cook all day to prepare the spicy rice and four fiery curries that compose *lampries*. They began grinding the spices while we guests were having breakfast, and it wasn't

52. Some interviewees in the magazine articles spoke about home cooking; some about gorging themselves on desserts when they were children; and some about picnic food. The writer wishes to add another example at the end of Paragraph 1 in order to show the range of experience covered. Which of the following sentences would best further the writer's purpose of capturing the diversity of the experiences described?

- F. Others recalled childhood feasts on ice cream or cake.
- G. When eating a seven-course meal, most diners must take care not to overindulge in appetizers.
- H. Grand restaurants make any meal memorable.
- J. Picnics—whether romantic or merely expedient—usually are composed of the same predictable dishes: bread, wine, cheese, pickles.

53. A. NO CHANGE

- B. (the name of which was Goodhue)
- C. (Goodhue was its name)
- D. OMIT the underlined portion.

54. F. NO CHANGE

- G. stands
- H. have stood
- J. were standing

55. The writer wishes to add another relevant example to Paragraph 2 without straying from the purpose of cataloguing a variety of meals, none of which can be more remarkable than the *lampries* meal described later in the essay. Which of the following sentences does that best?

- A. In my family, all Thanksgiving dinners are created equal.
- B. Surely, though, the most remarkable meal I've ever eaten, better and more exotic than all those curries, was the pure, elegant meal I had as part of a formal Japanese tea ceremony.
- C. There was the first meal I ever cooked all by myself, but oozy scrambled eggs and blackened toast are not memoir material.
- D. Dormitory food is, at best, bland and starchy.

56. F. NO CHANGE

- G. the most
- H. more
- J. OMIT the underlined portion.

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until late ⁵⁷afternoon that the innkeeper began cooking the curries. Then the cook layered them onto huge, green banana leaves, which she bundled up and tied into packets for the final baking.

[5]

All four rooms of the cottage were filled with the rich curry aromas of cinnamon, saffron, and mystery. The innkeeper brought out cold drinks and split a fresh pineapple. With two strokes of her machete, we ⁵⁸took our places, and with great ceremony the cook brought in the *lampries*—one fat packet for each of us. Inside was a mixture so unforgettably hot and exotic it shocked us, but we kept eating.

Items 59 and 60 pose questions about the essay as a whole.

59. The writer wishes to include the following sentence in the essay:

That evening, while standing out back listening to the eerie noises of the awakening night jungle, I peered in through the kitchen window to watch the innkeeper give each of the curries one final meditative stir.

That sentence will fit most smoothly and logically into Paragraph:

- A. 2, before the first sentence.
- B. 3, after the last sentence.
- C. 4, before the last sentence.
- D. 5, after the last sentence.

57. A. NO CHANGE
B. afternoon. That
C. afternoon, that
D. afternoon, thats when

58. F. NO CHANGE
G. pineapple. With two strokes of her machete we
H. pineapple, with two strokes of her machete, we
J. pineapple with two strokes of her machete. We

60. The writer wishes to make the details more lush and complete in order to help the reader virtually see, smell, and taste the *lampries*. Where should the writer place the following sentence?

“Are you hungry? Are you hungry?” chanted the innkeeper, and the dog barked with excitement as we cut the strings and undid the wrappings.

That sentence should be added to Paragraph:

- F. 3, before the second sentence.
- G. 3, after the third sentence.
- H. 4, after the last sentence.
- J. 5, before the last sentence.

PASSAGE V

Japanese Comic Books

When Americans think of comic books, they often think of children’s magazines starring Donald Duck and Superman. In Japan, however, comic books are among the most popular art forms,

GO ON TO THE NEXT PAGE.

which are being read by almost the entire
⁶¹
 population.

Japanese comic books, called *manga*,
⁶²
 look in appearance very different

⁶³
 then American comics. Whereas most American

comics are small ⁶⁴magazines. Japanese comics
 often resemble big-city telephone books, sometimes
 600 pages long. They are often published weekly—
⁶⁵
 there are also Korean and Chinese comics—and
 contain many serialized stories. The most popular
 stories are collected into permanent books.

⁶⁶
 Usually, American comic books are mostly
 about superheroes, a wide variety of stories appear in
 Japanese comics. There are adventure stories as in
 the U.S., but ⁶⁷they are drawn in a very unique
 style, unlike anything done in the American stories.

⁶⁸ In the U.S., comic books have

61. A. NO CHANGE
B. those being read in their entirety
C. read
D. OMIT the underlined portion.
62. F. NO CHANGE
G. look
H. look and appear
J. seem to appear to look
63. A. NO CHANGE
B. like
C. for
D. from
64. F. NO CHANGE
G. magazines,
H. for magazines.
J. magazines and
65. Which of the alternatives provides the contrast most appropriate and relevant to the essay?
A. NO CHANGE
B. newspapers are usually published daily—
C. American comics are usually issued on a monthly basis—
D. European comics are becoming more popular in America—
66. F. NO CHANGE
G. While
H. However,
J. OMIT the underlined portion.
67. Which of the alternative clauses would most effectively support the assertion made in the previous sentence about the variety of stories in Japanese comics?
A. NO CHANGE
B. there are also stories about sports, romance, work, history, school, and fashion.
C. American readers might find some of them difficult to understand, even in translation.
D. the most popular stories in Japanese comics are in the category of science fiction.
68. Which of the following would provide the best transition here, guiding the reader from the topic of the previous paragraph to the new topic of this paragraph?
F. Many people in Japan read comic books and also go to the movies.
G. As can be seen, there are many topics covered in Japanese comic books.
H. Very few females, young or old, read American comic books.
J. The varied subjects of Japanese *manga* attract a wide readership.

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traditionally been geared to either children or
teenage boys, and circulations rarely exceed

300,000. In Japan, therefore, males and females
of all ages read comics regularly, and there are
comics produced for distinct interests and age
groups. *Manga* can be found in waiting rooms,
barber shops, bus stations—almost anywhere.

We read on the train by businesspeople on the way
to work and by children coming home from school.

Circulations can easily soar into the millions.

In fact, 27 percent of all books and

magazines produced in Japan are comic
books. Comic book creators in Japan are superstars
comparable to America's most famous music and
television stars.

The comics industry is changing in the U.S.,
becoming much more like that of Japanese comics.

New kinds of stories for adults, as well as, children
are now being written, and some series are being
collected into permanent book format. However,
American comics have a long way to go before
they reach the level of artistic and cultural acceptance
that comics have achieved in Japan.

69. A. NO CHANGE
B. along
C. from
D. with

70. F. NO CHANGE
G. as well,
H. on the other hand,
J. consequently,

71. A. NO CHANGE
B. They
C. You
D. They are

72. F. NO CHANGE
G. magazines, produced in Japan,
H. magazines, produced in Japan
J. magazines produced in Japan,

73. A. NO CHANGE
B. that of the Japanese.
C. Japanese comics.
D. that of the Japanese comics industry.

74. F. NO CHANGE
G. adults as well as,
H. adults as well as
J. adults, as well as

GO ON TO THE NEXT PAGE.

1**1**

Item 75 poses a question about the essay as a whole.

75. Suppose the writer had been assigned to write a brief essay detailing some specific way in which Japan has been influenced by American culture. Would this essay successfully fulfill the assignment?
- A. No, because the essay suggests instead that the Japanese have been innovative leaders in developing the comic book as a popular art form.
 - B. No, because the essay does not give any specific examples of actual Japanese comic books.
 - C. Yes, because the U.S. does exert considerable influence on Japanese culture today, although the same is not true for the business world.
 - D. Yes, because the essay suggests that there are adventure stories in Japanese comic books, just as there are in American comic books.

END OF TEST 1

STOP! DO NOT TURN THE PAGE UNTIL TOLD TO DO SO.



MATHEMATICS TEST

60 Minutes—60 Questions

DIRECTIONS: Solve each problem, choose the correct answer, and then fill in the corresponding oval on your answer document.

Do not linger over problems that take too much time. Solve as many as you can; then return to the others in the time you have left for this test.

You are permitted to use a calculator on this test. You may use your calculator for any problems you choose,

but some of the problems may best be done without using a calculator.

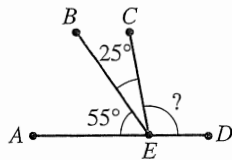
Note: Unless otherwise stated, all of the following should be assumed.

1. Illustrative figures are NOT necessarily drawn to scale.
2. Geometric figures lie in a plane.
3. The word *line* indicates a straight line.
4. The word *average* indicates arithmetic mean.

1. On a math test, 12 students earned an A. This number is exactly 25% of the total number of students in the class. How many students are in the class?

- A. 15
- B. 16
- C. 21
- D. 30
- E. 48

2. In the figure below, points A , E , and D are on the same line. What is the measure of $\angle CED$?



- F. 80°
- G. 90°
- H. 100°
- J. 110°
- K. 140°

3. What is the fifth term of the arithmetic sequence 8, 6, 4, ...?

- A. -2
- B. 0
- C. 4
- D. 8
- E. 16

4. What value of x solves the following proportion?

$$\frac{9}{6} = \frac{x}{8}$$

- F. $5\frac{1}{3}$
- G. $6\frac{3}{4}$
- H. $10\frac{1}{2}$
- J. 11
- K. 12

DO YOUR FIGURING HERE.

GO ON TO THE NEXT PAGE.



5. If point C bisects line segment $\overline{AB}$, then which of the following congruences must hold?

A. $\overline{CA} \cong \overline{AB}$
 B. $\overline{CB} \cong \overline{AB}$
 C. $\overline{AC} \cong \overline{BA}$
 D. $\overline{BC} \cong \overline{BA}$
 E. $\overline{AC} \cong \overline{CB}$

DO YOUR FIGURING HERE.

6. Three pieces of wire, each 2.8 meters long, are cut from the end of a wire 90 meters long. How many meters of wire are left?

F. 81.6
 G. 82.6
 H. 83.2
 J. 83.6
 K. 87.2

7. If $x = -3$, then $12 - 2(x + 1) = ?$

A. 4
 B. 7
 C. 16
 D. 19
 E. 20

8. $-|-10| - (-10) = ?$

F. -100
 G. -20
 H. 0
 J. 20
 K. 100

9. Umberto's mother expects an increase of 5% in her current annual salary of \$36,000. What would her new annual salary be?

A. \$36,005
 B. \$36,180
 C. \$37,800
 D. \$41,000
 E. \$54,000

10. If $x^3 = 20$ (and x is a real number), then x lies between which two consecutive integers?

F. 2 and 3
 G. 3 and 4
 H. 4 and 5
 J. 5 and 6
 K. 6 and 7

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11. If $78 - x = 234$, then $x = ?$

A. -312
 B. -156
 C. 3
 D. 156
 E. 312

DO YOUR FIGURING HERE.

12. In a laboratory experiment, Amoeba A lives 5 hours longer than Amoeba B, and Amoeba B lives twice as long as Amoeba C. If n is the lifespan of Amoeba C in hours, what is the lifespan of Amoeba A, in terms of n ?

F. $5 + 2n$
 G. $7 + n$
 H. $7n$
 J. $10n$
 K. $2(5 + n)$

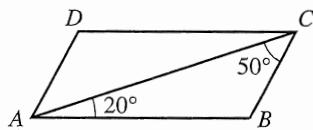
13. In the standard (x,y) coordinate plane, 3 corners of a rectangle are $(1,-1)$, $(-4,-1)$, and $(1,-4)$. Where is the rectangle's fourth corner?

A. $(1,4)$
 B. $(-1,4)$
 C. $(-1,1)$
 D. $(-1,-4)$
 E. $(-4,-4)$

14. Which of the following is a simplified form of $3x - 3y + 2x$?

F. $3(x - y + 2)$
 G. $(x - y)(3 + 2x)$
 H. $x(5 - 3y)$
 J. $x - 3y$
 K. $5x - 3y$

15. In the parallelogram below, what is the measure of $\angle DAC$?



A. 20°
 B. 30°
 C. 40°
 D. 50°
 E. 70°

GO ON TO THE NEXT PAGE.



16. What is the slope of any line parallel to the line $3x + 5y = 8$?

F. -3

G. $-\frac{3}{5}$

H. $\frac{3}{8}$

J. 3

K. 8

DO YOUR FIGURING HERE.

17. If $x > 0$ and $2x^2 - 5x - 12 = 0$, then $x =$?

A. $\frac{3}{2}$

B. 2

C. 4

D. 5

E. 12

18. The lengths of the sides of a triangle are 3, 8, and 9 inches. How many inches long is the shortest side of a similar triangle that has a perimeter of 60 inches?

F. 9

G. 11

H. 20

J. 24

K. 27

19. Jeans that normally sell for \$35.95 are on sale for 25% off. How much do they cost during the sale, to the nearest dollar?

A. \$ 9

B. \$11

C. \$26

D. \$27

E. \$29

20. Which of the following is a factored form of $4x^3y + 4xy^3$?

F. $4x^3y^3(y + x)$

G. $4xy(x^2 + y^2)$

H. $8xy(x^2 + y^2)$

J. $4x^3y^3$

K. $8x^4y^4$

21. If $2x - y = 6$, and $x + 4y = 12$, what is the value of y ?

A. -6

B. 0

C. 1

D. 2

E. 3

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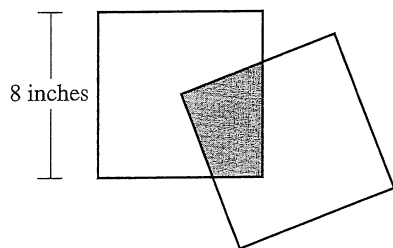


22. There are 16 ounces in a pound. If 1.5 pounds of hamburger costs \$2.88, what is the cost per ounce?

F. \$0.12
 G. \$0.18
 H. \$0.88
 J. \$1.92
 K. \$4.32

DO YOUR FIGURING HERE.

23. The 2 squares in the figure below have the same dimensions. The vertex of one square is at the center of the other square. What is the area of the shaded region, in square inches?



A. 4
 B. 8
 C. 16
 D. 32
 E. 64

24. A salesperson earns $(6h + 0.05t)$ dollars, where h is the number of hours worked, and t is the total amount of her sales. What does she earn for working 10 hours with \$230.80 in sales?

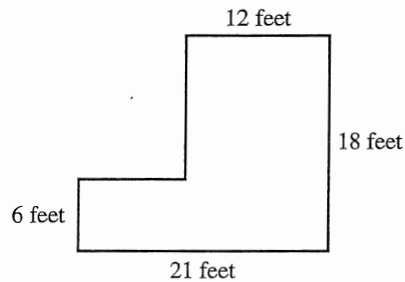
F. \$ 71.54
 G. \$ 74.54
 H. \$175.40
 J. \$251.80
 K. \$295.80

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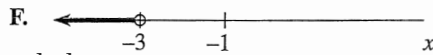
25. A floor has the dimensions shown below. How many square feet of carpeting are needed to cover the entire floor?

(Note: All angles are right angles.)

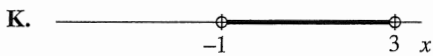
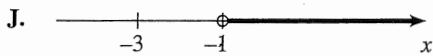
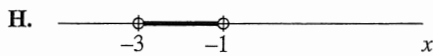
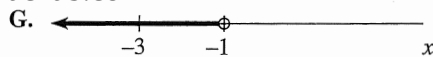


- A. 57
B. 72
C. 162
D. 270
E. 324

26. Which of the following is the graph of the solution set of $x - 1 < -2$?



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27. Which of the following is less than $\frac{3}{4}$?

- A. $\frac{5}{6}$
B. $\frac{5}{7}$
C. $\frac{6}{8}$
D. $\frac{8}{10}$
E. $\frac{10}{13}$

DO YOUR FIGURING HERE.

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2**2**

28. What is the area, in square meters, of a right triangle with sides of length 8 meters, 15 meters, and 17 meters?

DO YOUR FIGURING HERE.

- F. 40
- G. 60
- H. 68
- J. 120
- K. $127\frac{1}{2}$

29. When the marching band is arranged in rows of 4 people each, the last row is one person short. When it is arranged in rows of 5, the last row is still one person short. When arranged in rows of 6, the last row is *still* one person short. What is the least possible number of people in the marching band?

- A. 11
- B. 14
- C. 59
- D. 89
- E. 119

30. A triangle has sides of length 2.5 feet and 4 feet. Which of the following CANNOT be the length of the third side, in feet?

- F. 1
- G. 2
- H. 3
- J. 4
- K. 5

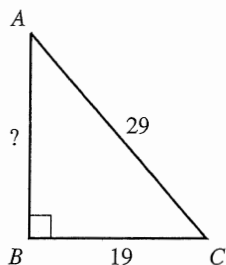
31. For all $a > 0$, $\frac{1}{a} + \frac{2}{3} = ?$

- A. $\frac{2}{3a}$
- B. $\frac{3}{3a}$
- C. $\frac{3+2a}{3a}$
- D. $\frac{3}{3+a}$
- E. $\frac{3+2a}{3+a}$

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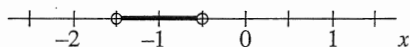


32. In the right triangle below, how long is side $\overline{AB}$?



DO YOUR FIGURING HERE.

- F. $\sqrt{29^2 - 19^2}$
 G. $\sqrt{29^2 + 19^2}$
 H. $29^2 - 19^2$
 J. $29^2 + 19^2$
 K. $29 - 19$
33. If the length of a square is increased by 1 inch and the width is increased by 2 inches, a rectangle is formed. If each side of the original square is x inches long, what is the area of the new rectangle, in square inches?
- A. $2x + 3$
 B. $4x + 6$
 C. $x^2 + 2$
 D. $x^2 + 3x + 2$
 E. $x^2 + 3x + 3$
34. If $\sin \alpha = \frac{12}{13}$, and $\cos \alpha = \frac{5}{13}$, then $\tan \alpha =$?
- F. $\frac{5}{12}$
 G. $\frac{7}{13}$
 H. $\frac{12}{5}$
 J. $\frac{17}{13}$
 K. $\frac{60}{13}$
35. Which of the following best describes the graph on the number line below?



- A. $|x| = -1$
 B. $|x| < 0.5$
 C. $-2 < x < 0$
 D. $-0.5 < x < -1.5$
 E. $-0.5 > x > -1.5$

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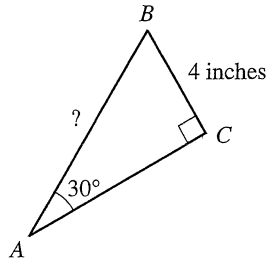


36. A performance was rated on a 3-point scale by an audience. A rating of 1 was given by 30% of the audience, a rating of 2 by 60%, and a rating of 3 by 10%. To the nearest tenth, what was the average of the ratings?

F. 1.2
G. 1.5
H. 1.8
J. 2.0
K. 2.2

DO YOUR FIGURING HERE.

37. In the right triangle below, if $\overline{BC}$ is 4 inches long, how many inches long is $\overline{AB}$?



A. 4
B. $4\sqrt{2}$
C. $4\sqrt{3}$
D. 6
E. 8

38. What is the largest possible product for 2 even integers whose sum is 38?

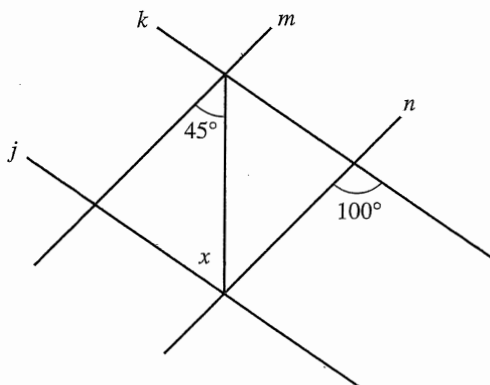
F. 72
G. 76
H. 136
J. 280
K. 360

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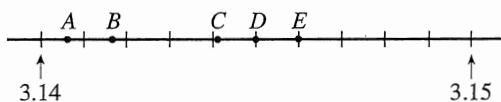


39. In the figure below, lines j and k are parallel, lines m and n are parallel, and the measures of 2 angles are shown. What is the measure of $\angle x$?

DO YOUR FIGURING HERE.



- A. 45°
 B. 55°
 C. 65°
 D. 75°
 E. 80°
40. In the (x,y) coordinate plane, what is the y -intercept of the line $6x - 2y = 6$?
- F. -3
 G. -2
 H. 0
 J. 3
 K. 6
41. Among the points graphed on the number line below, which is closest to π ?
- (Note: $\pi \approx 3.1415926$)



- A. A
 B. B
 C. C
 D. D
 E. E
42. For what value of a would the following system of equations have an infinite number of solutions?

$$\begin{aligned} 2x - y &= 6 \\ 8x - 4y &= 3a \end{aligned}$$

- F. 2
 G. 6
 H. 8
 J. 18
 K. 24

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43. The expression $(180 - x)$ is the degree measure of a nonzero acute angle if and only if:

A. $0 < x < 45$
B. $0 < x < 90$
C. $45 < x < 90$
D. $90 < x < 135$
E. $90 < x < 180$

DO YOUR FIGURING HERE.

44. If $x + y = -2$, and $x - y = -3$, then $x^2 - y^2 = ?$

F. 13
G. 6
H. 5
J. -5
K. -6

45. The sides of a triangle are 5, 12, and 13 inches long.
What is the angle between the 2 shortest sides?

A. 30°
B. 45°
C. 60°
D. 90°
E. 120°

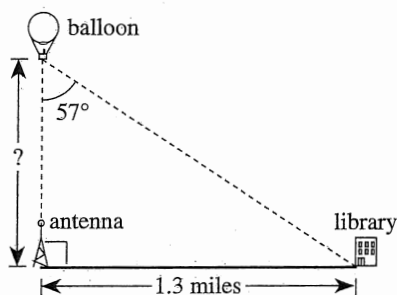
46. In the standard (x,y) coordinate plane, if the x -coordinate of each point on a line is 4 less than twice its y -coordinate, the slope of the line is:

F. -4
G. -2
H. $\frac{1}{2}$
J. 2
K. 4

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47. From a hot air balloon, the angle between a radio antenna straight below and the base of the library downtown is 57° , as shown below. If the distance between the radio antenna and the library is 1.3 miles, how many miles high is the balloon?



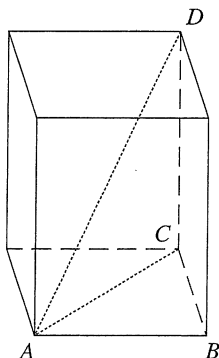
- A. $1.3 \sin 57^\circ$
 B. $1.3 \tan 57^\circ$
 C. $\frac{1.3}{\sin 57^\circ}$
 D. $\frac{1.3}{\cos 57^\circ}$
 E. $\frac{1.3}{\tan 57^\circ}$
48. Two numbers have a greatest common factor of 4 and a least common multiple of 24. Which of the following could be the pair of numbers?
 F. 4 and 8
 G. 4 and 12
 H. 8 and 12
 J. 8 and 24
 K. 12 and 24
49. Listed below are 5 functions, each denoted $g(x)$ and each involving a real number constant $c \geq 2$. If $f(x) = 2^x$, which of these 5 functions yields the greatest value for $f(g(x))$, for all $x > 1$?
 A. $g(x) = cx$
 B. $g(x) = \frac{c}{x}$
 C. $g(x) = \frac{x}{c}$
 D. $g(x) = x - c$
 E. $g(x) = \log_c x$

DO YOUR FIGURING HERE.

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50. Line segments $\overline{AB}$, $\overline{BC}$, and $\overline{CD}$, which represent the 3 dimensions of the rectangular box shown below, have lengths of 4 inches, 3 inches, and 5 inches, respectively. What is the sine of $\angle DAC$?



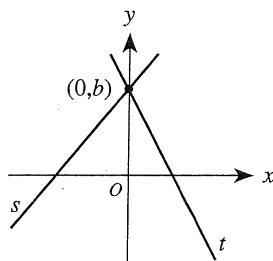
- F. 1
G. $\frac{4}{5}$
H. $\frac{\sqrt{2}}{2}$
J. $\frac{3}{5}$
K. $\frac{3\sqrt{2}}{10}$

DO YOUR FIGURING HERE.

51. A certain circle has an area of π square inches. How many inches long is its radius?

- A. $\frac{1}{2}$
B. 1
C. 2
D. $\frac{\pi}{2}$
E. π

52. The equation of line s below is $y = mx + b$. Which of the following could be an equation for line t ?



- F. $y = 2mx$
G. $y = 2mx + b$
H. $y = 2mx - b$
J. $y = -2mx + b$
K. $y = -2mx - b$

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53. The equation $x^2 - 10x + k = 0$ has only one solution for x . What is the value of k ?

A. 0
B. 5
C. 10
D. 20
E. 25

DO YOUR FIGURING HERE.

54. In the standard (x,y) coordinate plane, what is the slope of the line that passes through the origin and the point $\left(\frac{1}{2}, \frac{2}{3}\right)$?

F. $\frac{1}{2}$
G. $\frac{1}{3}$
H. $\frac{2}{3}$
J. $\frac{3}{4}$
K. $\frac{4}{3}$

55. If A , B , and C are real numbers, and if $ABC = 1$, which of the following conditions *must* be true?

A. AB is equal to $\frac{1}{C}$
B. A , B , and C must all be positive
C. Either $A = 1$, $B = 1$, or $C = 1$
D. Either $A = 0$, $B = 0$, or $C = 0$
E. Either $A < 1$, $B < 1$, or $C < 1$

56. The length of a rectangle is $(x + 5)$ units and its width is $(x + 7)$ units. Which of the following expresses the remaining area of the rectangle, in square units, if a square, x units in length, is removed from the interior of the rectangle?

F. 12
G. 35
H. $2x + 12$
J. $12x + 35$
K. $x^2 + 12x + 35$

GO ON TO THE NEXT PAGE.



57. What is the smallest positive value for x where $y = \sin 2x$ reaches its maximum?

A. $\frac{\pi}{4}$
B. π
C. $\frac{3\pi}{2}$
D. 2π
E. $\frac{5\pi}{2}$

DO YOUR FIGURING HERE.

58. In the standard (x,y) coordinate plane, if the distance between points $(8,a)$ and $(a,1)$ is 5 coordinate units, which of the following could be the value of a ?

F. 5
G. 3
H. -3
J. -4
K. -5

59. Martin has an empty bag and puts in 3 red marbles. He now wants to put in enough green marbles so the probability of drawing a red marble at random from the bag is $\frac{1}{4}$. How many green marbles should he put in?

A. 1
B. 3
C. 5
D. 9
E. 12

60. How many different integer values of n satisfy the inequality $\frac{1}{13} < \frac{2}{n} < \frac{1}{10}$?

F. 1
G. 2
H. 3
J. 4
K. 5

END OF TEST 2

STOP! DO NOT TURN THE PAGE UNTIL TOLD TO DO SO.

DO NOT RETURN TO THE PREVIOUS TEST.

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READING TEST

35 Minutes—40 Questions

DIRECTIONS: There are four passages in this test. Each passage is followed by several questions. After reading a passage, choose the best answer to each question and fill in the corresponding oval on your answer document. You may refer to the passages as often as necessary.

Passage I

PROSE FICTION: This passage is adapted from Paule Marshall's short story "Reena" (©1983 by The Feminist Press).

We met—Reena and myself—at the funeral of her aunt who had been my godmother and whom I had also called aunt, Aunt Vi, and loved, for she and her house had been, respectively, a source of understanding and a place of calm for me as a child. Reena entered the church where the funeral service was being held as though she, not the minister, were coming to officiate, sat down among the immediate family up front, and turned to inspect those behind her. I saw her face then.

It was a good copy of the original. The familiar mold was there, that is, and the configuration of bone beneath the skin was the same despite the slight fleshiness I had never seen there before, her features had even retained their distinctive touches: the positive set to her mouth, the assertive lift to her nose, the same insistent, unsettling eyes which when she was angry became as black as her skin—and this was total, unnerving, and very beautiful. Yet something had happened to her face. It was different despite its sameness. Aging even while it remained enviably young. Time had sketched in, very lightly, the evidence of the twenty years.

Her real name had been Doreen, a standard for girls among West Indians (her mother, like my parents, was from Barbados), but she had changed it to Reena on her twelfth birthday—"As a present to myself"—and had enforced the change on her family by refusing to answer to the old name. "Reena. With two e's!" she would say and imprint those e's on your mind with the indelible black of her eyes and a thin threatening finger that was like a quill.

She and I had not been friends through our own choice. Rather, our mothers, who had known each other since childhood, had forced the relationship. And from the beginning, I had been at a disadvantage. For Reena, as early as the age of twelve, had had a quality that was unique, superior, and therefore dangerous. She seemed defined, even then, all of a piece, the raw edges of her adolescence smoothed over; indeed, she seemed to have escaped adolescence altogether and made one dazzling leap from childhood into the very arena of adult life.

At thirteen, for instance, she was reading Zola, Hauptmann, Steinbeck, while I was still in the thrall of the Little Minister and Lorna Doone. When I could only barely conceive of the world beyond Brooklyn, she was talking of the Civil War in Spain, lynchings in the South, Hitler in Poland—and talking with the outrage and passion of a revolutionary. I would try, I remember, to console myself with the thought that she was really an adult masquerading as a child, which meant that I could not possibly be her match.

For her part, Reena put up with me and was, by turns, patronizing and impatient. I merely served as the audience before whom she rehearsed her ideas and the yardstick by which she measured her worldliness and knowledge.

"Do you realize that this stupid country supplied Japan with the scrap iron to make the weapons she's now using against it?" she had shouted at me once.

I had not known that.

Just as she overwhelmed me, she overwhelmed her family, with the result that despite a half dozen brothers and sisters who consumed quantities of bread and jam whenever they visited us, she behaved like an only child and got away with it. Her father, a gentle man with skin the color of dried tobacco and with the nose Reena had inherited jutting out like a crag from his nondescript face, had come from Georgia and was always making jokes about having married a foreigner—Reena's mother being from the West Indies. When not joking, he seemed slightly bewildered by his large family and so in awe of Reena that he avoided her. Reena's mother, a small, dry, formidably black woman, was less a person to me than the abstract principle of force, power, energy. She was alternately strict and indulgent with Reena and, despite the inconsistency, surprisingly effective.

1. Of the persons mentioned in the passage, which of the following had the greatest positive effect on the narrator as a child?

- A. Reena's minister
- B. Reena's father
- C. Aunt Vi's godmother
- D. Aunt Vi

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2. In order to ensure that her family would call her Reena, and not Doreen, Reena would:
- I. point at them threateningly.
 - II. start crying loudly.
 - III. shout and stamp her feet.
 - IV. stare meaningfully.
- F. I and II only
G. I and IV only
H. II and IV only
J. I, II, and IV only
3. It can reasonably be inferred from the passage that Reena's mother, as compared with Reena's father, was a:
- A. more strict and much funnier parent.
 - B. more retiring and less authoritative parent.
 - C. more forceful and effective parent.
 - D. less argumentative and more gentle parent.
4. Reena's talking about which of the following subjects intimidated the narrator?
- I. Hitler in Poland
 - II. The Civil War in Spain
 - III. The thrall of the Little Minister
- F. I only
G. II only
H. III only
J. I and II only
5. As it is described in the first paragraph, Reena's entrance into the church suggests that Reena is a woman who:
- A. is quite confident.
 - B. is used to officiating at funerals.
 - C. is deeply unhappy.
 - D. has changed remarkably.
6. Reena apparently had the sort of character that her father found it necessary to:
- F. discipline her severely.
 - G. keep her at a distance.
 - H. praise her constantly.
 - J. humor her endlessly.
7. The narrator's point of view is that of:
- A. a child.
 - B. an adolescent.
 - C. a psychologist.
 - D. an adult.
8. The statement that Reena had a half dozen brothers and sisters yet "behaved like an only child and got away with it" (lines 64–65) supports the narrator's feeling that Reena:
- F. was completely and utterly selfish.
 - G. had been her best friend for years.
 - H. did not like her brothers and sisters.
 - J. could overwhelm just about anyone.
9. According to the narrator, adolescence is a stage usually characterized by:
- A. raw edges.
 - B. abstract principles.
 - C. dazzling leaps.
 - D. impatient patronizing.
10. The fifth paragraph (lines 52–56) suggests that Reena's relationship with the narrator was primarily characterized by:
- F. Reena's patience with the narrator.
 - G. Reena's exploitation of the narrator.
 - H. the narrator's devotion to Reena.
 - J. the narrator's increasing worldliness.

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Passage II

SOCIAL SCIENCE: This passage is adapted from Jack Weatherford's *Indian Givers: How the Indians of America Transformed the World* (©1988 by Jack McIver Weatherford).

Egalitarian democracy and liberty as we know them today owe little to Europe. They are not Greco-Roman derivatives somehow revived by the French in the eighteenth century. They entered modern western thought as American Indian notions translated into European language and culture.

In language, custom, religion, and written law, the Spaniards descended directly from ancient Rome, yet they brought nothing resembling a democratic tradition with them to America. The French and Dutch who settled parts of North America also settled many other parts of the world that did not become democratic. Democracy did not spring up on French-speaking Haiti any more than in Southern Africa, where the British and Dutch settled about the same time that they settled in North America.

Even the Netherlands and Britain, the two showcases for European democracy, had difficulty grafting democracy onto monarchical and aristocratic systems soaked in the strong traditions of class privilege. During the reign of George III of Great Britain, while the United States was fighting for its independence, only one person in twenty could vote in England. And in Ireland no Catholic could hold office or vote. In their centuries of struggle to suppress the Irish, the British possibly encumbered their own democratic development.

American anglophiles occasionally point to the signing of the Magna Carta by King John on the battlefield of Runnymede in 1215 as the start of civil liberties and democracy in the English-speaking world. This document, however, merely moved slightly away from monarchy and toward oligarchy by increasing the power of the aristocracy. It continued the traditional European vacillation between government by a single strong ruler and by an oligarchic class. An oligarchy is not an incipient democracy, and a step away from monarchy does not necessarily mean a step toward democracy. In the same tradition, the election of the pope by a college of cardinals did not make the Vatican into a democratic institution, nor did the Holy Roman Empire become a democracy merely because a congress of aristocrats elected the emperor.

When the Dutch built colonies in America, power in their homeland rested securely in the hands of the aristocracy and the burghers, who composed only a quarter of the population. A city such as Amsterdam fell under the rule of a council of thirty-six men, none of whom was elected; instead each council member inherited his office and held it until death.

Henry Steele Commager wrote that during the Enlightenment "Europe was ruled by the wellborn, the rich, the privileged, by those who held their places by

divine favor, inheritance, prescription, or purchase." The philosophers and thinkers of the Enlightenment became quite complacent and self-congratulatory because the "enlightened despots" such as Catherine of Russia and Frederick of Prussia read widely and showed literary inclinations. Too many philosophers became court pets and because of that believed that Europe was moving toward enlightened democracy. As Commager explained it, Europe only imagined the Enlightenment, but America enacted it. This Enlightenment grew as much from its roots in Indian culture as from any other source.

When Americans try to trace their democratic heritage back through the writings of French and English political thinkers of the Enlightenment, they often forget that these people's thoughts were heavily shaped by the democratic traditions and the state of nature of the American Indians. The concept of the "noble savage" derived largely from writings about the American Indians, and even though the picture grew romanticized and distorted, the writers were only romanticizing and distorting something that really did exist. The Indians did live in a fairly democratic condition, they were egalitarian, and they did live in greater harmony with nature.

The modern notions of democracy based on egalitarian principles and a federated government of overlapping powers arose from the unique blend of European and Indian political ideas and institutions along the Atlantic coast between 1607 and 1776. Modern democracy as we know it today is as much the legacy of the American Indians, particularly the Iroquois and the Algonquians, as it is of the British settlers, of French political theory, or of all the failed efforts of the Greeks and Romans.

The discovery of new forms of political life in America freed the imaginations of Old World thinkers to envision utopias, socialism, communism, anarchism, and dozens of other social forms. Scarcely any political theory or movement of the last three centuries has not shown the impact of this great political awakening that the Indians provoked among the Europeans.

11. According to the passage, two Native American peoples who contributed greatly to the development of modern democracy were the:
- A. Iroquois and the Cherokee.
 - B. Iroquois and the Algonquians.
 - C. Algonquians and the Seminoles.
 - D. Cherokee and the Cheyenne.

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12. The author of the passage would most likely agree with which of the following statements?
- F. European political thinkers of the sixteenth century created the notion of a completely egalitarian society.
 - G. The efforts of the Spaniards to create a democratic society in the New World failed due to the unfavorable climate of the New World.
 - H. American Indians generally are not given as much credit as they deserve with regard to their contribution to modern democratic political theory.
 - J. The roots of modern democracy can be traced directly back to the Holy Roman Empire.
13. Historian Henry Steele Commager's belief that "Europe only imagined the Enlightenment, but America enacted it" (lines 62–63) refers to the idea, presented in the passage, that:
- A. European political thinkers wrote a great deal about democracy and liberty, but democracy and liberty did not really manifest themselves until European and Native American political ideas met in the New World.
 - B. European political thinkers lived utopian lives that prevented them from seeing the monarchical excesses of European society.
 - C. the Dutch and Spanish political thinkers had a history of democratic traditions, but they were not able to translate their ideas into a workable democracy in America.
 - D. Native Americans, when introduced to the democratic ideals of European political thinkers, readily adopted the Europeans' political philosophies.
14. One of the main ideas of the passage is that:
- F. democracy and liberty are political ideas derived primarily from the Greeks and Romans of the ancient world.
 - G. the French and the Dutch who settled in America were the primary sources of democracy in the New World.
 - H. modern democracy evolved from the interaction of Native American and European political thought in colonial America.
 - J. Native Americans were initially opposed to the democratic traditions that the Europeans brought to the New World.
15. It can be inferred from the sixth paragraph (lines 51–65) that historian Henry Steele Commager would agree with the statement that, during the Enlightenment, Europe was mainly ruled by:
- A. a democratic majority.
 - B. a college of cardinals.
 - C. the aristocratic class.
 - D. the intellectual elite.
16. The passage argues that at the time of European contact with Native Americans in the 1600s, the political systems of Native Americans could best be characterized as being:
- F. essentially nonexistent.
 - G. ruled by a few Indian chiefs who were similar to Europe's "enlightened despots."
 - H. a monarchical system of government.
 - J. fairly democratic and egalitarian.
17. The passage specifies that the law of which of the following countries descended directly from that of ancient Rome?
- A. Britain
 - B. France
 - C. The Netherlands
 - D. Spain
18. According to the fourth paragraph (lines 28–43), the signing of the Magna Carta:
- I. increased the power of the English aristocracy.
 - II. decreased the power of the English monarchy.
 - III. created the first truly democratic government in England.
- F. I only
 - G. I and II only
 - H. I and III only
 - J. II and III only
19. According to the passage, the attitude of some philosophers of the Enlightenment toward European monarchs and their governments was often:
- A. not critical enough, because the philosophers were on too friendly terms with the monarchs.
 - B. not critical enough, because the philosophers needed to justify European expansion in North America.
 - C. too critical, because the philosophers personally disliked the monarchs.
 - D. too critical, because the philosophers didn't understand Greco-Roman ideas well enough to develop sound theories.
20. According to the passage, at the same time they settled in North America, the British and the Dutch also settled in:
- I. Haiti.
 - II. South Africa.
 - III. Greece.
- F. I only
 - G. II only
 - H. I and II only
 - J. I and III only

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Passage III

HUMANITIES: This passage is adapted from Annie Dillard's *The Writing Life* (©1989 by Annie Dillard).

When you write, you lay out a line of words. The line of words is a miner's pick, a woodcarver's gouge, a surgeon's probe. You wield it, and it digs a path you follow. Soon you find yourself deep in new territory. Is it a dead end, or have you located the real subject? You will know tomorrow, or this time next year.

You make the path boldly and follow it fearfully. You go where the path leads. At the end of the path, you find a box canyon. You hammer out reports, dispatch bulletins.

The writing has changed, in your hands, and in a twinkling, from an expression of your notions to an epistemological tool. The new place interests you because it is not clear. You attend. In your humility, you lay down the words carefully, watching all the angles. Now the earlier writing looks soft and careless. Process is nothing; erase your tracks. The path is not the work. I hope your tracks have grown over; I hope birds ate the crumbs; I hope you will toss it all and not look back.

The line of words is a hammer. You hammer against the walls of your house. You tap the walls, lightly, everywhere. After giving many years' attention to these things, you know what to listen for. Some of the walls are bearing walls; they have to stay, or everything will fall down. Other walls can go with impunity; you can hear the difference. Unfortunately, it is often the bearing wall that has to go. It cannot be helped. There is only one solution, which appalls you, but there it is. Knock it out. Duck.

Courage utterly opposes the bold hope that this is such fine stuff the work needs it, or the world. Courage, exhausted, stands on bare reality: this writing weakens the work. You must demolish the work and start over. You can save some of the sentences, like bricks. It will be a miracle if you can save some of the paragraphs, no matter how excellent in themselves or hard-won. You can waste a year worrying about it, or you can get it over with now. (Are you a woman, or a mouse?)

The part you must jettison is not only the best-written part; it is also, oddly, that part which was to have been the very point. It is the original key passage, the passage on which the rest was to hang, and from which you yourself drew the courage to begin.

Putting a book together is interesting and exhilarating. It is sufficiently difficult and complex that it engages all your intelligence. It is life at its most free. Your freedom as a writer is not freedom of expression in the sense of wild blurting; you may not let it rip. It is life at its most free, if you are fortunate enough to be able to try it, because you select your materials, invent your task, and pace yourself.

The obverse of this freedom, of course, is that your work is so meaningless, so fully for yourself alone, and so worthless to the world, that no one except you cares whether you do it well, or ever. You are free to make several thousand close judgment calls a day. Your freedom is a by-product of your days' triviality.

Here is a fairly sober version of what happens in the small room between the writer and the work itself. It is similar to what happens between a painter and a canvas.

First you shape the vision of what the projected work of art will be. The vision, I stress, is no marvelous thing: it is the work's intellectual structure and aesthetic surface. It is a chip of mind, a pleasing intellectual object. It is a vision of the work, not of the world. It is a glowing thing, a blurred thing of beauty. Its structure is at once luminous and translucent; you can see the world through it.

Many aspects of the work are still uncertain, of course; you know that. You know that if you proceed you will change things and learn things, that the form will grow under your hands and develop new and richer lights. But that change will not alter the vision or its deep structures; it will only enrich it. You know that, and you are right.

But you are wrong if you think that in the actual writing, or in the actual painting, you are filling in the vision. You cannot fill in the vision. You cannot even bring the vision to light. You are wrong if you think you can in any way take the vision and tame it to the page. The page is jealous and tyrannical; the page is made of time and matter; the page always wins. The vision is not so much destroyed, exactly, as it is, by the time you have finished, forgotten.

21. As it is used in line 47, the word *engages* most nearly means:
- A. demands.
 - B. defeats.
 - C. envisions.
 - D. ensures.
22. The author compares the interaction between writers and their work to that in all of the following occupations EXCEPT:
- F. surgeon.
 - G. miner.
 - H. painter.
 - J. musician.

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23. The author suggests that the best-written part of a piece of writing is often, ironically, the part of a piece of writing that the writer:
- A. finds most painful.
 - B. must throw away.
 - C. feels is most dramatic.
 - D. produced in a wild burst.
24. Which of the following best states the main point of the passage?
- F. Writers need to be aggressive and intellectual.
 - G. The path is really the same thing as the work.
 - H. Writing is a humbling and transforming experience.
 - J. In writing, it is crucial that you consider your audience.
25. The main emphasis of the third paragraph (lines 11–20) regarding the nature of the act of writing is on:
- A. why writers need to learn humility.
 - B. keeping the line of words from being altered.
 - C. how a writer's perception of her work changes.
 - D. how writing expresses notions of the self.
26. As it is used in line 9, the phrase *hammer out* most nearly means:
- F. break.
 - G. write.
 - H. erase.
 - J. remove.
27. An analogy made in the passage is that sentences are to writing as:
- A. courage is to bare reality.
 - B. bearing walls are to vision.
 - C. bricks are to building.
 - D. painting is to freedom.
28. The author claims that putting a book together is life at its most free because:
- F. you select your own materials, task, and pace.
 - G. you can fully express your inner self.
 - H. nothing is more intellectually demanding.
 - J. you create something valued by the entire world.
29. The author of the passage describes the vision as:
- I. a chip of mind.
 - II. the by-product of your day's triviality.
 - III. a glowing thing.
- A. II only
 - B. III only
 - C. I and II only
 - D. I and III only
30. The author asserts that it will be a miracle if, during the course of revision, the writer is able to salvage:
- F. some of the bricks.
 - G. any of the words.
 - H. some of the paragraphs.
 - J. all of the path.

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Passage IV

NATURAL SCIENCE: This passage is adapted from Frank Close, Michael Marten, and Christine Sutton's *The Particle Explosion* (©1987 by Frank Close, Michael Marten, and Christine Sutton).

The detector is a kind of ultimate microscope, which records what happens when a [subatomic] particle strikes another particle, either in a fixed target such as a lump of metal or a chamber filled with a gas or liquid, or in an on-coming beam in a collider. The 1950s and 60s were the age of the bubble chamber, so called because electrically charged particles moving through it produce trails of tiny bubbles in the liquid filling the chamber. [But today most] experiments are based on electronic detectors.

Detectors rarely record *all* the particle collisions that occur in a particular experiment. Usually collisions occur thousands of times a second and no equipment can respond quickly enough to record all the associated data. Moreover, many of the collisions may reveal mundane 'events' that are relatively well understood. So the experimenters often define beforehand the types of event that may reveal the particles they are trying to find, and program the detector accordingly. This is what a major part of the electronics in a detector is all about. The electronics form a filter system, which decides within a split second whether a collision has produced the kind of event that the experimenters have defined as interesting and which should therefore be recorded by the computer. Of the thousands of collisions per second, only one may actually be recorded. One of the advantages of this approach is its flexibility: the filter system can always be reprogrammed to select different types of event.

Often, computer graphics enable the events to be displayed on computer monitors as images, which help the physicists to discover whether their detector is functioning in the correct way and to interpret complex or novel events.

Imaging has always played an important role in particle physics. In earlier days, much of the data was actually recorded in photographic form—in pictures of tracks through cloud chambers and bubble chambers, or even directly in the emulsion of special photographic film. Many of these images have a peculiar aesthetic appeal, resembling abstract art. Even at the subatomic level nature presents images of itself that reflect our own imaginings.

The essential clue to understanding the images of particle physics is that they show the *tracks* of the particles, not the particles themselves. What a pion, for instance, really looks like remains a mystery, but its passage through a substance—solid, liquid, or gas—can be recorded. Particle physicists have become as adept at interpreting the types of track left by different particles as the American Indians were at interpreting the tracks of an enemy.

A number of simple clues immediately narrows down the possibilities. For instance, many detectors are based around a magnet. This is because the tracks of electrically-charged particles are bent in a magnetic field. A curving track is the signature of a charged particle. And if you know the direction of the magnetic field, then the way that the track curves—to left or right, say—tells you whether the particle is positively or negatively charged. The radius of curvature is also important, and depends on the particle's velocity and mass. Electrons, for instance, which are very light-weight particles, can curve so much in a magnetic field that their tracks form tight little spirals.

Most of the subatomic zoo of particles have brief lives, less than a billionth of a second. But this is often long enough for the particle to leave a measurable track. Relatively long-lived particles leave long tracks, which can pass right through a detector. Shorter-lived particles, on the other hand, usually decay visibly, giving birth to two or more new particles. These decays are often easily identified in images: a single track turns into several tracks.

Neutral particles present more of a headache to experimenters. Particles without an electric charge leave no tracks in a detector, so their presence can be deduced only from their interactions or their decay products. If you see two tracks starting at a common point, apparently arising from nowhere, you can be almost certain that this is where a neutral particle has decayed into two charged particles.

Our perception of nature has deepened not only because the accelerators have increased in power, but also because the detection techniques have grown more sophisticated. The quality of particle imagery and the range of information it provides have both improved over the years.

31. The main idea of the passage is that:

- A. most particle collisions are "mundane" events.
- B. bubble chambers were constructed to capture high-energy particles.
- C. the technology for detecting particle images is improving.
- D. the detection of particle images has direct application to the study of nuclear energy.

32. The passage states that magnets affect atomic particles by:

- F. influencing the direction particles travel.
- G. turning particles into negatively charged electrons.
- H. increasing the life of particles.
- J. causing positive and negative particles to collide.

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33. The passage states that which of the following particles leaves a long track?
- A. A positively charged particle
 - B. A negatively charged particle
 - C. A short-lived particle
 - D. A long-lived particle
34. As it is used in line 46, the word *pion* precisely refers to:
- F. an image.
 - G. a track.
 - H. a particle.
 - J. a molecule.
35. According to the passage, which of the following CANNOT be tracked electronically by experimenters?
- A. Electrically charged particles
 - B. Pion particles
 - C. Negatively charged particles
 - D. Neutral particles
36. Which of the following statements would the authors most likely agree with?
- F. Most tracking of electrically charged particles is difficult and inaccurate.
 - G. Tracking of electrically charged particles is still primitive because of unclear photographs.
 - H. Short-lived particles are easier to track than long-lived particles.
 - J. Electrically charged particles can be tracked with the right equipment and careful observation.
37. What, according to the passage, is one effect of charged particles passing through a bubble chamber?
- A. Collisions of the particles as they are stopped by the bubbles
 - B. Computer images that can be greatly enhanced
 - C. Photographs of the actual particles
 - D. Patterns of tiny bubbles in the liquid filling the chamber
38. The passage suggests that the greatest difference between experiments done with a bubble chamber and those done with electronic detectors is that:
- F. bubble chambers are much better at tracking the particles.
 - G. electronic detectors can track pions.
 - H. electronic detectors are more selective of the particle events.
 - J. electronic detectors can photograph the particles themselves.
39. How does the analogy likening the detector to the microscope function in the passage?
- A. It suggests that the detector, like the microscope, reveals to scientists a part of reality not easily seen.
 - B. It presents the differences and similarities in the way a detector works compared to a microscope.
 - C. It proves that all instruments are ultimately the same in the way that they function in a laboratory.
 - D. It introduces the argument in the passage that all detectors, whether microscope, bubble chamber, or collider, present images that resemble abstract art.
40. What is the main idea of the second paragraph (lines 11–29)?
- F. Even the best detectors still miss most of the important collisions in an experiment.
 - G. New technology allows scientists to select the collisions they want to record.
 - H. Despite the new technology, detectors still record mostly mundane events.
 - J. Scientists can now use computers to record virtually all the collisions in an experiment.

END OF TEST 3

STOP! DO NOT TURN THE PAGE UNTIL TOLD TO DO SO.

DO NOT RETURN TO A PREVIOUS TEST.



SCIENCE REASONING TEST

35 Minutes—40 Questions

DIRECTIONS: There are seven passages in this test. Each passage is followed by several questions. After reading a passage, choose the best answer to each question and fill in the corresponding oval on your answer document. You may refer to the passages as often as necessary.

You are NOT permitted to use a calculator on this test.

Passage I

Bacteria reproduce by a process in which a single cell divides into two cells. The average time required for bacteria to divide and their population to double is called the *generation time*. Table 1 presents the generation time for a variety of bacteria at a given temperature.

Bacterial population growth occurs in a series of distinct steps referred to as a *bacterial growth curve*. It consists of 4 phases, which reflect changes in the cells' environment and metabolism over time. See Figure 1.

Table 1			
Bacterium	Growth medium	Temperature (°C)	Generation time (min)
<i>Clostridium botulinum</i>	glucose broth	37	35
<i>Escherichia coli</i>	glucose broth	37	17
<i>Lactobacillus acidophilus</i>	milk	37	66
<i>Mycobacterium tuberculosis</i>	synthetic medium	37	792
<i>Pseudomonas aeruginosa</i>	glucose broth	37	31
<i>Shigella dysenteriae</i>	milk	37	23
<i>Staphylococcus aureus</i>	glucose broth	37	32
<i>Streptococcus lactis</i>	lactose broth	30	48
<i>Streptococcus lactis</i>	glucose milk	37	26
<i>Streptococcus lactis</i>	peptone milk	37	37
<i>Streptococcus pneumoniae</i>	glucose broth	37	30
<i>Xanthomonas campestris</i>	glucose broth	25	74

Table adapted from P. L. Altman and D. S. Dittmer, eds., *Biology Data Book*. ©1972 by Pergamon Press PLC.

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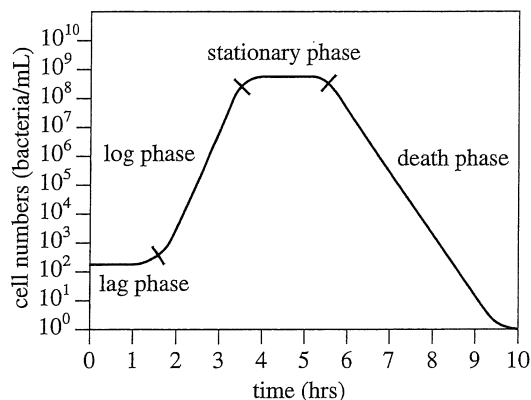


Figure 1

- On the basis of the data presented in Table 1, if *Streptococcus lactis* was placed in a test tube containing growth medium at 37°C, one would predict its generation time to most likely be:
 - less than 10 min.
 - between 10 and 15 min.
 - between 15 and 25 min.
 - between 25 and 40 min.
- The bacterial population increases most rapidly during which of the following phases?
 - Lag phase
 - Log phase
 - Stationary phase
 - Death phase
- Based on the data presented in Table 1, which of the following bacteria growing in glucose broth took the longest time to double its population?
 - Pseudomonas aeruginosa*
 - Staphylococcus aureus*
 - Streptococcus pneumoniae*
 - Xanthomonas campestris*
- Samples of growth medium containing milk were inoculated with the microorganisms depicted in Table 1 and put in an environmental chamber at 37°C. Which of the following bacteria would take closest to 1 hour to double its population?
 - Lactobacillus acidophilus*
 - Mycobacterium tuberculosis*
 - Shigella dysenteriae*
 - Streptococcus lactis*
- Which of the following hypotheses about bacterial populations is supported by the data presented in Figure 1?
 - Bacteria populations change at a constant rate throughout all the growth phases.
 - Bacteria begin to increase immediately after transfer to a new growth medium.
 - Bacteria begin to decrease immediately after transfer to a new growth medium.
 - Bacteria require a period of adjustment before they begin to reproduce in a new growth medium.

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Passage II

Scientists noted an increase in plant growth in a lake. Increased growth of lake weeds and algae is usually the result of an increased input of nutrients, especially phosphates and nitrates. Nitrates are easily carried by water moving through the soil (*groundwater*) or streams. Phosphates can attach to soil or stream sediment particles. To determine the primary source of nutrients entering the lake, scientists conducted the following experiments.

Experiment 1

Scientists deduced that one source of phosphates and nitrates was seepage from wastewater systems buried in the soil near houses adjacent to the lake. Sampling wells were placed in locations where samples of groundwater, flowing from houses toward the lake, could be obtained daily. The results are presented in Table 1.

Table 1		
Date	Phosphate concentration (mg/L)	Nitrate concentration (mg/L)
House 1		
May 2	7.4	17.2
May 3	8.4	17.9
May 4	8.0	18.3
May 5	7.7	17.5
May 6	7.2	21.7
House 2		
May 2	9.1	22.8
May 3	9.7	25.1
May 4	11.8	22.5
May 5	9.1	21.3
May 6	8.8	18.2

Experiment 2

Scientists suspected that another source of nutrients was the runoff from nearby farm lands where fertilizers were applied. Water and suspended-sediment samples were obtained from two streams that flowed into the lake. These streams intercept surface runoff from the farm lands during rainfall and snowmelt. The results are depicted in Table 2.

Table 2			
Date	Suspended sediment concentration (mg/L)	Phosphate concentration (mg/L)	Nitrate concentration (mg/L)
Stream 1			
May 2	14.1	8.6	37.4
May 3	16.4	10.3	36.3
May 4	477.2	45.8	38.9
May 5	1,080.9	90.2	61.1
May 6	568.6	50.3	58.2
Stream 2			
May 2	8.3	7.6	10.7
May 3	15.5	15.2	24.8
May 4	25.1	27.3	27.4
May 5	17.2	16.9	21.6
May 6	8.3	10.4	11.1

6. How do the designs of Experiments 1 and 2 differ in terms of the sampling procedure?
 - F. In Experiment 1, sampling was performed daily, whereas in Experiment 2, sampling was performed weekly.
 - G. In Experiment 1, groundwater was sampled, whereas in Experiment 2, stream water was sampled.
 - H. In Experiment 1, suspended sediment concentration was sampled, whereas in Experiment 2, suspended sediment concentration was not sampled.
 - J. In Experiment 1, only nitrate concentration was sampled, whereas in Experiment 2, only phosphate concentration was sampled.
7. In order to obtain more information about the relationship between phosphates, nitrates, and plant growth, which of the following procedures should be performed next?
 - A. Studying how the lake weeds grow in water maintained at different temperatures
 - B. Growing algae in water samples containing several different phosphate and nitrate concentrations
 - C. Adding large amounts of phosphates and nitrates to the soil of House 1
 - D. Decreasing the amount of irrigation used by local farmers

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8. Scientists suspected that fertilizers applied to farm lands also contaminated groundwater, which in turn increased nutrient input into the lake. In order to best test this hypothesis, which of the following should the scientists do next?
- F. Sample water from three different streams
 - G. Measure the increase in algae and plant growth in the lake
 - H. Increase the amount of fertilizer applied to nearby farm lands
 - J. Vary the amount of fertilizer applied to nearby farm lands and sample groundwater flowing toward the lake in these fields
9. What was the scientists' hypothesis concerning lake plant growth in Experiment 1?
- A. Wastewater from houses increases lake plant growth.
 - B. Runoff from farm lands decreases lake plant growth.
 - C. Rain falling in the vicinity of the houses and lake decreases lake plant growth.
 - D. Lake sediments release nitrates and phosphates into areas where plants are least abundant.
10. Given the results of Experiments 1 and 2, all of the following measures would reduce the input of phosphorus and nitrogen into the lake EXCEPT:
- F. increasing the number of houses surrounding the lake.
 - G. limiting surface runoff and erosion on upstream farm lands.
 - H. limiting fertilizer application on upstream farm lands.
 - J. installing pipes to carry wastewater to a central treatment facility.
11. As phosphates move farther from their source, they are more likely to be adsorbed by the soil (removed from the water). Which of the following would most likely be the approximate phosphate concentration for House 1 on May 3 if the sampling well were closer to the wastewater system of the house?
- A. 6.0 mg/L
 - B. 7.0 mg/L
 - C. 8.0 mg/L
 - D. 9.0 mg/L

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Passage III

Motor vehicle exhaust is a significant source of several air pollutants, including the nitrogen oxides, NO_2 and NO . Scientists performed two experiments to investigate the levels and behavior of these pollutants and naturally occurring gases in the atmosphere near busy roadways.

Experiment 1

Scientists studied how NO_2 levels vary with vehicle use. They measured NO_2 levels hourly for a 24-hour period, 10 meters (m) downwind from six roadways. Each roadway had a combination of a different speed limit and *vehicle usage* (approximate number of vehicles per day). An average NO_2 value for each roadway was calculated in parts per billion (ppb). The results are in Table 1.

Table 1		
Roadway speed limit (km/hr)	Vehicle usage (vehicles/day)	Average NO_2 level (ppb)
60	10,000	5
	20,000	9
	30,000	13
100	10,000	8
	20,000	13
	30,000	22

Experiment 2

Next, the levels of NO_2 and NO were measured at 0, 50, and 100 m downwind from the 100 km/hr roadway which averaged 30,000 vehicles per day. The level of NO_2 decreased from 30 ppb at 0 m to 17 ppb at 100 m. The level of NO decreased from 150 ppb at 0 m to 42 ppb at 100 m.

(Note: The levels of NO_2 and NO that would be found far from pollution sources are 15 ppb and 35 ppb, respectively.)

Experiment 3

Ozone (O_3) is a naturally occurring gas in the atmosphere. Levels of ozone, in parts per million (ppm) were taken at various distances downwind from the 100 km/hr roadway which averaged 30,000 vehicles per day. The results are in Table 2.

(Note: The naturally occurring ozone concentration is 0.12 ppm.)

Table 2	
Distance from roadway (m)	Ozone level (ppm)
0	0.0075
50	0.02
100	0.04
150	0.075
200	0.09

12. Which of the following factors was varied in Experiment 3 ?
 - F. Background concentration of NO_2
 - G. Background level of ozone
 - H. Distance from roadway
 - J. Speed limit
13. According to the experimental results, one way to reduce levels of NO_2 in an area would be to:
 - A. reduce the levels of naturally occurring ozone near roadways.
 - B. lower speed limits on the roadways.
 - C. raise speed limits on the roadways.
 - D. require installation of ozone filters in motor vehicle exhaust systems.
14. According to the experimental results, if one compared ozone levels near a major highway to those in a remote wilderness location, ozone levels:
 - F. near the highway would be higher than at the wilderness location.
 - G. near the highway would be lower than at the wilderness location.
 - H. near the highway would be the same as those in the wilderness location.
 - J. would be detectable only near the highway.

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15. Carbon monoxide (CO) is another pollutant associated with motor vehicle exhaust. If carbon monoxide behaves like the nitrogen oxides in the experiments, one would expect that carbon monoxide levels:
- A. would decrease over time.
 - B. would stay the same over time.
 - C. are higher near roadways than farther away from them.
 - D. are lower near roadways than farther away from them.
16. According to the results of the experiments, as distance from the roadway increases:
- F. NO₂ and ozone levels both increase.
 - G. NO₂ levels increase and ozone levels decrease.
 - H. NO₂ levels increase and ozone levels stay the same.
 - J. NO₂ levels decrease and ozone levels increase.
17. A certain roadway has a speed limit of 100 km/hr and an average vehicle usage of 100,000 vehicles per day. One would predict that 10 m downwind from this roadway NO₂ levels are:
- A. less than 8 ppb.
 - B. between 8 and 13 ppb.
 - C. between 13 and 22 ppb.
 - D. above 22 ppb.

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Passage IV

The following experiments were done to study factors that affect the precession of a top. *Precession* is the revolution of a top around an imaginary line perpendicular to the surface at the point of contact (see Figure 1).

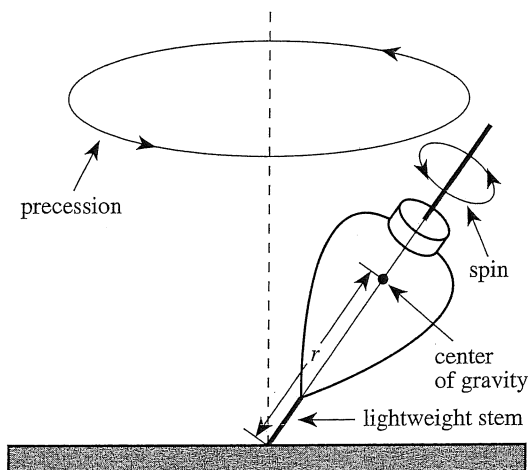


Figure 1

Experiment 1

An electric motor was attached to a top and used to give the top a known spin rate or number of revolutions per minute (rpm). Once the desired spin rate was achieved, the motor was removed and the number of precessions per minute was counted. The process was repeated using different spin rates. The results are recorded in Table 1.

Table 1	
Spin rate (rpm)	Precession rate (rpm)
350	16
500	11
700	8
1,100	5

Experiment 2

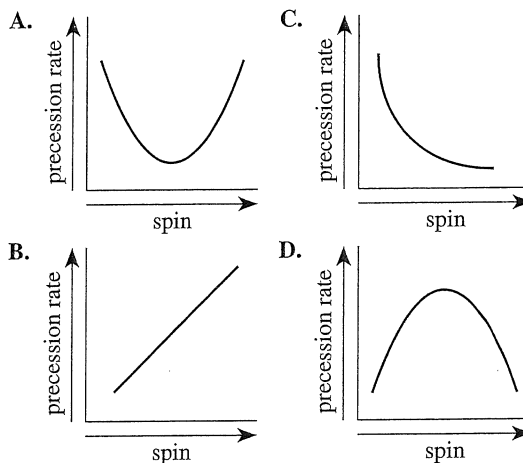
The distance from the surface to the top's center of gravity (r) was varied by changing the length of the lightweight stem on the top from Experiment 1. The electric motor was used to generate the same spin rate in each trial. The measured precession rate is given in Table 2.

Table 2	
r (inches)	Precession rate (rpm)
2	7.5
3	11
4	15
5	19

Experiment 3

A scientist took the top from Experiment 1 to the Moon. It was found that for a given spin rate and stem length, the precession rate was approximately one-sixth that on Earth. For example, a precession rate of 12 rpm on Earth would be approximately 2 rpm on the Moon.

18. Based on the results of Experiment 2, one can conclude that the precession rate of a top increases as the stem:
- F. decreases in length.
 - G. increases in length.
 - H. remains the same length.
 - J. doubles in mass.
19. Which of the following graphs best represents the change in precession rate with increasing spin rate as shown in Experiment 1?



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20. The scientist conducting Experiment 3 hypothesizes that precession rate is related to gravity. The best way to confirm this hypothesis would be to repeat the experiment on:
- F. several different planets with varying gravities.
 - G. one other planet with the same gravity as Earth.
 - H. a planet without an atmosphere.
 - J. Earth using various spin rates.
21. If the spin rate used in Experiment 2 was 500 rpm, what is most likely the value of r in Experiment 1 ?
- A. 3 inches
 - B. 4 inches
 - C. 5 inches
 - D. 6 inches
22. If the techniques of Experiment 1 had not been perfected first, how would this have affected Experiment 2 ?
- F. The mass of the top would have been a factor.
 - G. The spin rate might not have been the same in each trial.
 - H. The top's mass may have redistributed itself.
 - J. The top's shape may have changed.
23. How would one best investigate the effect of a top's mass on the precession rate if the spin rate is constant?
- A. Use tops of different colors
 - B. Use tops that are the same size, have the same shape, and are made from different metals
 - C. Try different tops and test them on both the Earth and Moon
 - D. Try tops with equal mass but stems of different lengths

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Passage V

Lightning is a visible electrical emission that often occurs in a cloud that has positive charges at the top and negative charges at the base (bottom). Two theories attempt to explain the mechanisms by which thunderstorm clouds become electrically charged and produce lightning.

Gravitational Theory

As shown in Stage I of Figure 1, in a *mature thunderstorm cloud* (a cloud beginning to precipitate), most precipitation particles, which may include raindrops, snow, or ice crystals, are initially *neutral* (no charge). As the larger particles (water droplets or ice crystals) in the upper regions of such a cloud fall due to gravity, they pass and rub against the smaller particles (see Stage II). The friction between them generates static electricity and places a negative charge on the larger particles as they settle to the cloud base. The smaller particles, which become positively charged through this process, rise to the top of the cloud (see Stage III). This separation of the positive and negative charges within the cloud is necessary for lightning formation.

Convective Theory

In this theory, the wind circulation within thunderstorm clouds is believed to cause the charge separation. Near the ground, some air molecules are positively charged. As *updrafts* (vertically rising currents of air) from the surface initiate the development of thunderstorm clouds, they also carry these positive charges aloft into the cloud (see Stage I of Figure 2). Initially, the entire cloud has a positive charge (Stage II). Negatively charged particles in the atmosphere are attracted toward the positively charged cloud and form a layer of negative charge around the edges of the cloud (Stage III). Strong downdrafts form as precipitation begins to fall and the thunderstorm reaches its mature state (Stage IV). These downdrafts strip the layer of negative charge from the cloud edge and carry it downward toward the base. The net effect of the convective circulation is to carry positively charged particles from the atmosphere to the cloud top and deposit negatively charged particles from the cloud's edge at the base of the cloud. This process continues through the remainder of the cloud's life cycle.

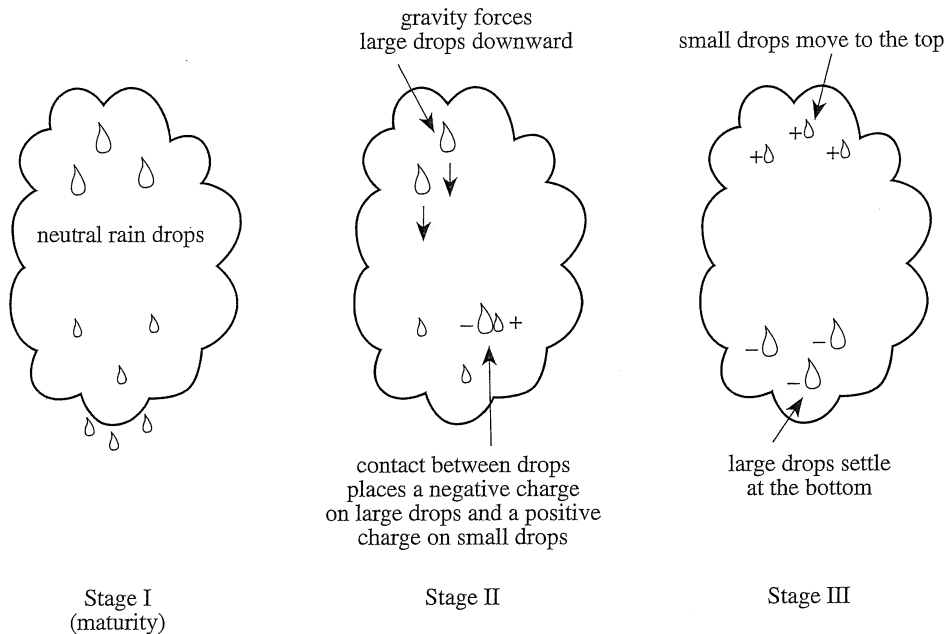


Figure 1

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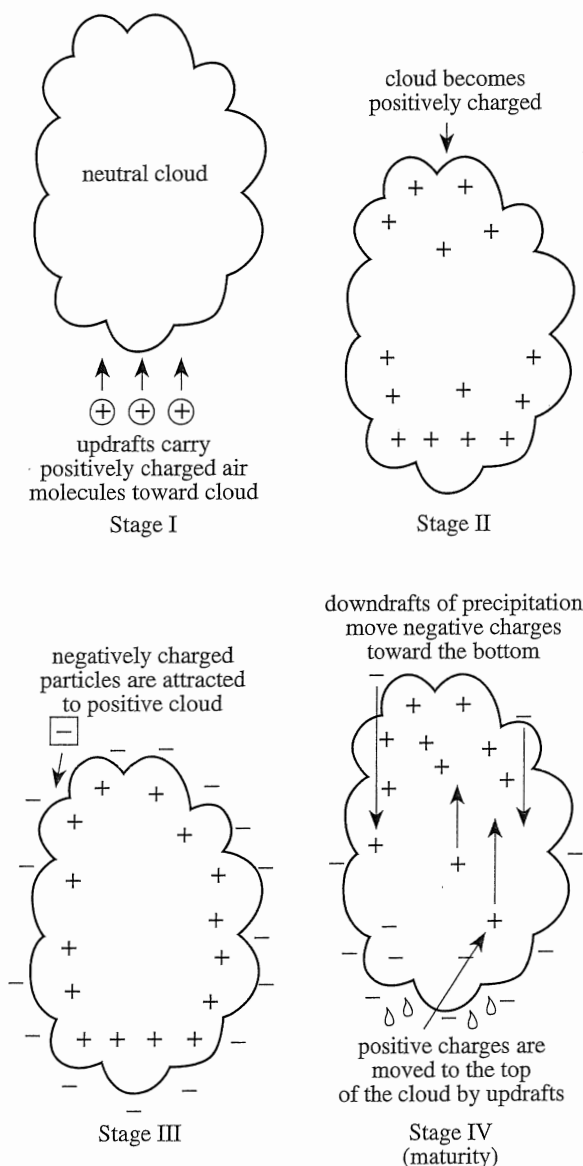


Figure 2

24. Which of the following represents an example of a cloud in which supporters of both theories would agree lightning would NOT occur?
- F. A cloud with large precipitation particles colliding with other smaller precipitation particles
 - G. A cloud with updrafts and downdrafts moving charged particles from the atmosphere
 - H. A cloud that only contains neutral particles
 - J. A cloud with positive charges at the top of the cloud and negative charges at the base of the cloud
25. About which of the following points do the two theories differ?
- A. Height of thunderstorm clouds
 - B. Initial location of charges in the cloud
 - C. Whether precipitation forms
 - D. Amount of rainfall produced
26. According to the Gravitational Theory, the concentration of negative charge at the cloud base is greatest:
- F. when the raindrops inside the cloud are large.
 - G. when the raindrops inside the cloud are small.
 - H. when no rain is falling inside the cloud.
 - J. after a lightning strike.
27. A research aircraft flies through the center of a mature thunderstorm cloud at a constant altitude and finds that most of the particles in that region are negatively charged. Both theories would agree that:
- A. no circulation is present to carry positively charged particles upward from the ground.
 - B. most of the positively charged particles are suspended above this region.
 - C. strong updrafts are occurring only at levels above this region.
 - D. the entire cloud is negatively charged.
28. Florida experiences more thunderstorms than any other state. The Convective Theory would suggest that these storms most likely result from which of the following?
- F. A high negative charge in the air just above the ground
 - G. A greater attractive force between positive and negative charges in this region
 - H. A high frequency of updrafts over Florida
 - J. A low frequency of updrafts over Florida
29. If a cloud consisted entirely of ice crystals, which of the following statements about this cloud would support the Gravitational Theory?
- A. Only the ice crystals with positive charge would fall and create a separation of charge.
 - B. Positively charged ice crystals would attract negatively charged ice crystals from the atmosphere.
 - C. The larger ice crystals would fall more rapidly than the smaller ice crystals and become negatively charged.
 - D. Downdrafts cause the ice crystals to fall and therefore gain negative charge.
30. Lightning usually occurs when electrical current flows from regions of negative charge to regions with positive charge. Based on the Convective Theory, which lightning path is NOT possible?
- F. Within a cloud from base to top
 - G. From the base of one cloud to the top of another
 - H. Cloud top to the ground
 - J. Cloud base to a positively charged region of the atmosphere

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Passage VI

An atom has a *nucleus*, which consists of *protons* (positively charged particles) and *neutrons* (uncharged particles), surrounded by one or more *electrons* (negatively charged particles) which move in *orbits* (see Figure 1).

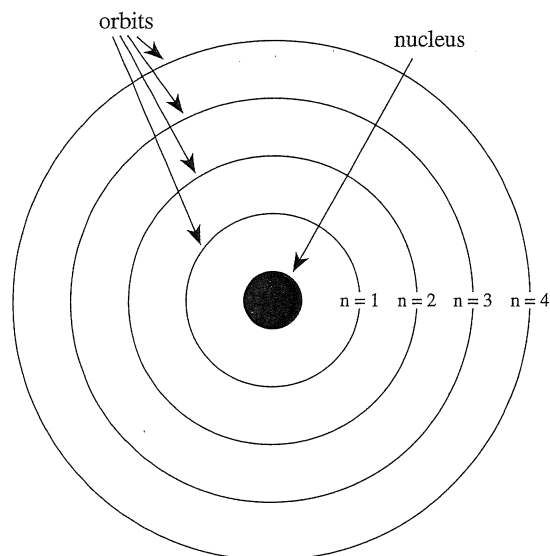


Figure 1

Note: Drawing is NOT to scale.

Table 1					
Element	Z	n	$r(\times 10^{-8} \text{ cm})$	I (eV)	E (eV)
H	1	4	8.5	0.85	12.7
H	1	3	4.8	1.5	12.1
H	1	2	2.1	3.4	10.2
He ⁺	2	4	4.2	3.4	51.0
He ⁺	2	3	2.4	6.0	48.4
He ⁺	2	2	1.1	13.6	40.9
Li ⁺²	3	4	2.8	7.7	115.0
Li ⁺²	3	3	1.6	13.6	109.0
Li ⁺²	3	2	0.7	30.5	91.9
Be ⁺³	4	4	2.1	13.6	204.0
Be ⁺³	4	3	1.2	24.2	194.0
Be ⁺³	4	2	0.5	54.2	163.0

31. For a given value of Z, the data indicate that as n increases, I:
- A. increases.
 - B. increases, then decreases.
 - C. decreases.
 - D. decreases, then increases.
32. According to Table 1, for element H, $n = 5$, the most likely value for I would be:
- F. 0.54 eV.
 - G. 0.90 eV.
 - H. 2.7 eV.
 - J. 3.4 eV.
33. According to the information provided in the passage, an atom has a net positive charge when it has more:
- A. neutrons than protons.
 - B. protons than neutrons.
 - C. protons than electrons.
 - D. electrons than neutrons.

Table 1 contains data for several neutral and charged atoms containing only one electron. Table 1 includes the number of protons (Z); the identification of the initial orbit (n); the radius of the orbit (r); the energy (I, in electron volts, or eV) to remove an electron from the atom; and the energy (E) of a photon emitted when an electron falls from its initial orbit to the orbit where $n = 1$.

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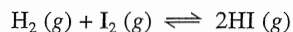
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34. The hypothesis that for a given value of n , r decreases as net positive charge increases, is supported by the data for:
- F. H and He^+ only.
 - G. H and Li^{+2} only.
 - H. H, He^+ , and Li^{+2} only.
 - J. H, He^+ , Li^{+2} , and Be^{+3} .
35. The most energetic photons will be emitted when an electron falls from orbit:
- A. $n = 2$ to $n = 1$ in He^+ .
 - B. $n = 2$ to $n = 1$ in Be^{+3} .
 - C. $n = 4$ to $n = 1$ in H.
 - D. $n = 4$ to $n = 1$ in Be^{+3} .

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Passage VII

Consider the following reaction:



At *chemical equilibrium*, the formation of hydrogen iodide gas (HI) occurs at the same rate as the formation of hydrogen (H_2) and iodine (I_2) gases.

At equilibrium, the equilibrium constant, (K_{eq}), is the ratio of the square of the HI concentration ($[\text{HI}]$) to the product of the concentrations of H_2 and I_2 . The table below shows the data collected while varying either the temperature or the initial reactant concentrations.

Table 1								
Trial	Temperature (°C)	K_{eq}	Initial $[\text{H}_2]$ (mol/L*)	Initial $[\text{I}_2]$ (mol/L)	Initial $[\text{HI}]$ (mol/L)	Final $[\text{H}_2]$ (mol/L)	Final $[\text{I}_2]$ (mol/L)	Final $[\text{HI}]$ (mol/L)
1	360	66.0	0.50	0.50	0	0.099	0.099	0.802
2	380	61.9	0.50	0.50	0	0.100	0.100	0.800
3	400	57.7	0.50	0.50	0	0.106	0.106	0.788
4	420	53.7	0.50	0.50	0	0.109	0.109	0.782
5	440	50.5	0.50	0.50	0	0.110	0.110	0.780
6	460	46.8	0.50	0.50	0	0.113	0.113	0.774
7	400	57.7	1.00	1.00	0	0.208	0.208	1.584
8	420	53.7	1.00	1.00	0	0.214	0.214	1.572
9	400	57.7	1.00	0.50	0	0.529	0.029	0.942
10	400	57.7	0.50	1.00	0	0.029	0.529	0.942
11	400	57.7	0.75	0.25	0	0.508	0.008	0.484
12	400	57.7	0.25	0.75	0	0.008	0.508	0.484
13	400	57.7	0.15	0.85	0	0.002	0.702	0.296

* mol/L = moles/liter

Table adapted from Ralph H. Petrucci, *General Chemistry-Principles and Modern Applications*. ©1985 by Macmillan Publishing Company.

6. Which of the following pairs of trials in the table shows that reversing the initial concentrations of H_2 and I_2 at the same temperature gives the same final concentration of HI?
- F. Trials 1 and 2
 G. Trials 7 and 8
 H. Trials 8 and 11
 J. Trials 9 and 10
37. Based on the information in the table, as the initial concentrations of H_2 and I_2 are both doubled, the final concentration of HI:
- A. decreases by one-half.
 B. decreases by one-fourth.
 C. increases by one-fourth.
 D. increases by two times.

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38. According to the table, increasing the reaction temperature, while keeping the total initial concentration of reactants constant, would result in which, if any, of the following concentration changes?
- F. A decrease in the initial concentration of H_2 and I_2
 - G. A decrease in the final concentration of HI
 - H. An increase in the initial concentration of H_2 and I_2
 - J. No changes in the final concentration of the product
39. Based on the data in the table, which of the following statements best represents the relationship that exists between the K_{eq} of a reaction and the temperature of that reaction?
- A. The K_{eq} value decreases as the temperature is increased.
 - B. The K_{eq} value decreases then levels off as the temperature is increased.
 - C. The K_{eq} value stays the same as the temperature is increased.
 - D. The K_{eq} value increases as the temperature is increased.
40. Based on the information in the table, if the scientist had repeated the experiment at 400°C and started with 2.0 mol/L each of H_2 and I_2 , the final concentration of HI would be closest to:
- F. 0.35 mol/L .
 - G. 0.70 mol/L .
 - H. 1.50 mol/L .
 - J. 3.00 mol/L .

END OF TEST 4

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1	A	36	H	1	E	31	C	1	D	1	D
2	G	37	D	2	H	32	F	2	G	2	G
3	D	38	J	3	B	33	D	3	C	3	D
4	F	39	C	4	K	34	H	4	J	4	F
5	A	40	F	5	E	35	E	5	A	5	D
6	J	41	D	6	F	36	H	6	G	6	G
7	A	42	F	7	C	37	E	7	D	7	B
8	G	43	B	8	H	38	K	8	J	8	J
9	C	44	H	9	C	39	B	9	A	9	A
10	H	45	B	10	F	40	F	10	G	10	F
11	B	46	F	11	B	41	B	11	B	11	D
12	J	47	C	12	F	42	H	12	H	12	H
13	C	48	J	13	E	43	E	13	A	13	B
14	J	49	A	14	K	44	G	14	H	14	G
15	D	50	H	15	D	45	D	15	C	15	C
16	G	51	B	16	G	46	H	16	J	16	J
17	A	52	F	17	C	47	E	17	D	17	D
18	H	53	D	18	F	48	H	18	G	18	G
19	D	54	G	19	D	49	A	19	A	19	C
20	H	55	C	20	G	50	H	20	G	20	F
21	D	56	G	21	D	51	B	21	A	21	A
22	G	57	A	22	F	52	J	22	J	22	G
23	D	58	J	23	C	53	E	23	B	23	B
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30	H	65	C	30	F	60	K	30	H	30	H
31	C	66	G					31	C	31	C
32	G	67	B					32	F	32	F
33	D	68	J					33	D	33	C
34	H	69	A					34	H	34	J
35	A	70	H					35	D	35	D
		71	D					36	J	36	J
		72	F					37	D	37	D
		73	B					38	H	38	G
		74	H					39	A	39	A
		75	A					40	G	40	J